Human fall detection using accelerometer and gyroscope sensors in unconstrained smartphone positions

Abstract

This study explored several methods for detecting body falls based on the data captured by the sensors (accelerometer and gyroscope) built in a smartphone carried by a person. The data for this study were collected by recording many sample units from each of the following human activity categories: stand-fall, walk-fall, stand-jump, stand-sit, stand, and walk. Several time-series data captured by the sensors were used as human motion features. One of the challenges of this study was the existence of human body motions whose features resembled those of body falls. In addition, unfixed smartphone positioning made human body falls harder to detect and can lead to high rate of misclassification (not detected as fall). This incident can caused serious bone fracture or even death if the person not handled as immediately as possible because of misclassification. To address this problem, we modified Resultant Acceleration and ∠ Y formulas to address the problem of unconstrained smartphone positions. We proposed to combine five methods such as AGVeSR, Alim, ∠α, GyroReDi, and AGPeak to build a robust detector model to reduce the misclassification. The experiment results showed that the accuracy of the combination of both sensors (accelerometer and gyroscope) outperformed the accuracy of accelerometer only by more than 15%. The decision fusion that used voting involving five methods could boost the accuracy rate by up to 4.15%. © BEIESP.

Author keywords

Accelerometer; Gyroscope; Human fall detection; Unconstrained smartphone positions